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Olist insight: delivery, satisfaction & prediction in brazilian e-commerce

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Data Warehouse, OLAP analysis and predictive modelling on 100K+ real orders from Olist (2016–2018)

MOTIVATION

Olist connects Brazilian SMEs to major marketplaces. With 100K+ orders across 27 states, delays and negative reviews directly impact seller reputation and revenue. This project builds a Data Warehouse to uncover what drives customer dissatisfaction and predict it before it happens.

BIG QUESTIONS

Descriptive

Sales, delays and ratings by region and category

Which regions and product categories have the highest volume, longest delays and lowest review scores and how have these metrics evolved over time?

Diagnostic

Delay and freight cost → negative review

Is there a statistically significant association between delivery delay, freight cost and negative reviews and what is the relative weight of each factor?

Predictive

Predict negative review before delivery

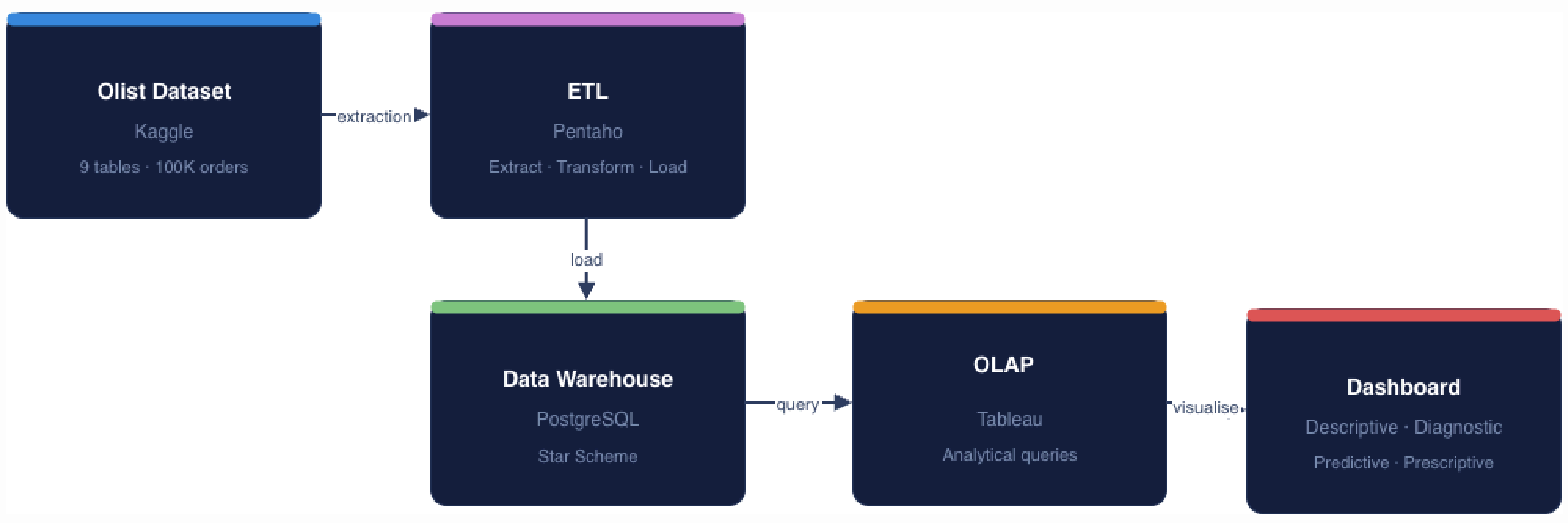
Can we predict a 1-2 star review using only pre-delivery attributes (category, region, seller, freight)? Which variables have the highest predictive power?

Prescriptive

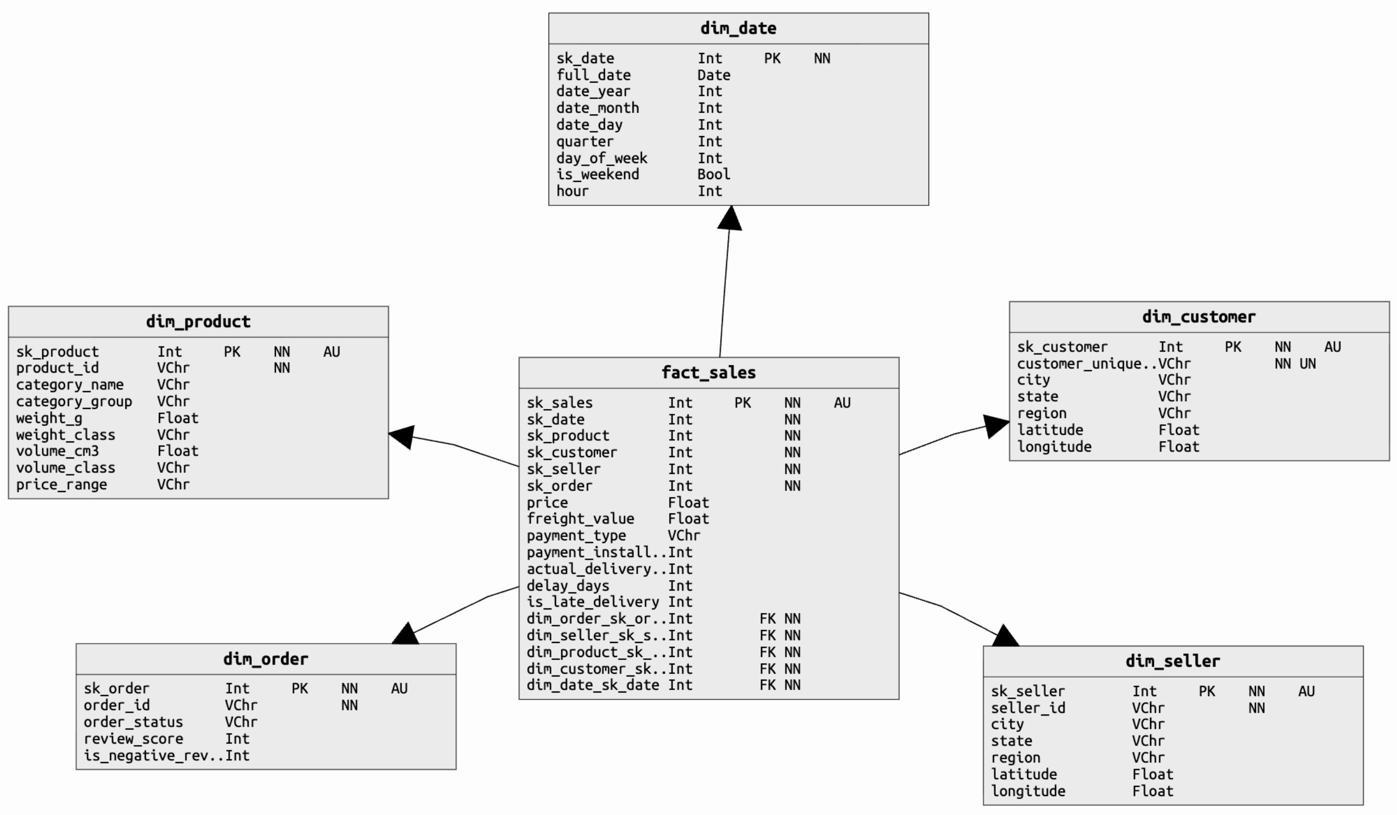
Proactive delay alerts

Can we estimate delivery time and automatically trigger a preventive action for high-risk orders minimising the impact on customer satisfaction?

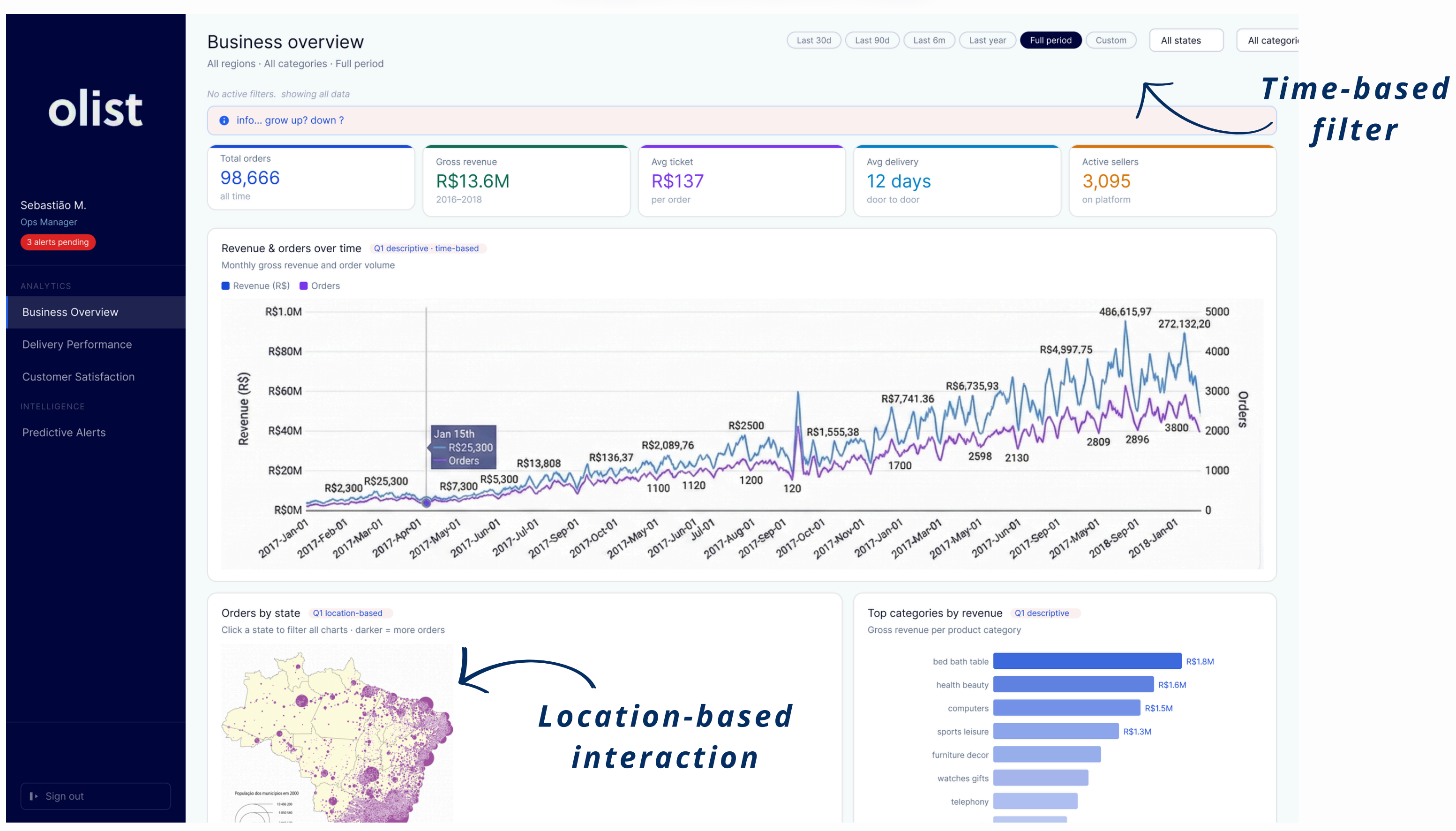
WORKFLOW

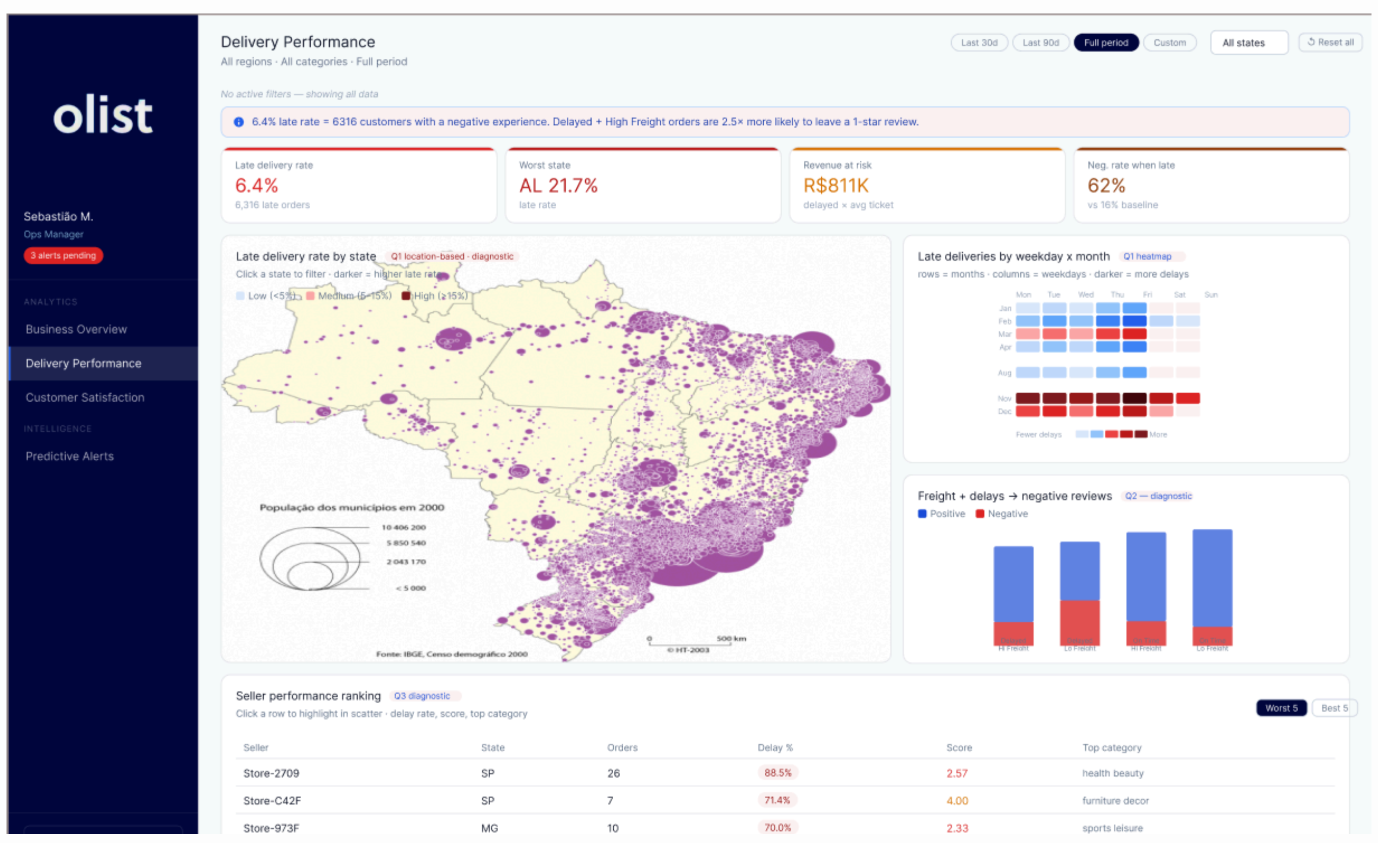


STAR SCHEME



USER INTERFACE





EXPECTED RESULTS

Delay and satisfaction map

Identify regions and seller profiles with the highest late delivery rates and their direct impact on review scores.

Freight and delay impact

Quantify the relative contribution of freight cost and delay days to the probability of a negative review.

Negative review classifier

A model to predict 1-2 star reviews before delivery, trained on pre-delivery order features.

Proactive alert system

For high-risk orders, automatically recommend a preventive action before the customer is affected.